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ELECTRONIC DEVICE FOR TRANSFERRING HEAT FROM HEATING ELEMENT TO COOLING DUCT

Abstract

An electronic device being excellent in disassembly workability at the time of repair or the like while sufficiently ensuring thermal conductivity for transferring heat from a heating element to a cooling duct. A circuit board has at least one heating element that generates heat by energization, a cooling duct is disposed to face the circuit board and through which air for cooling the heating element passes, a heat dissipation filler is provided in contact with a surface of the cooling duct facing the circuit board and having thermal conductivity for transferring heat from the heating element to the cooling duct, a flexible member is provided between the circuit board and the heat dissipation filler and having flexibility, and a positioning unit which positions the flexible member at a position where contact between the circuit board and the heat dissipation filler can be prevented.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an electronic device.

Description of the Related Art

[0002] In recent image pickup apparatuses, image quality has been improved by, for example, an increase in resolution, an increase in frame rate, and the like of recorded video. In such an image pickup apparatus, a processing load on an electronic component such as an IC chip mounted on a control circuit board increases with the improvement in image quality, and as a result, the electronic component significantly generates heat. Since heat generated in the electronic component may cause performance degradation, failure, or the like of the image pickup apparatus, a cooling structure for cooling the electronic component is provided. As the cooling structure, for example, a liquid heat dissipation member called heat dissipation grease or heat conductive grease may be used (see, for example, Japanese Laid-Open Patent Publication (kokai) No. 2015-115417 and Japanese Laid-Open Patent Publication (kokai) No. 2001-77569). In general, the liquid heat dissipation member has excellent heat reduction properties for reducing contact thermal resistance or the like generated at a contact surface between components as compared with a solid heat dissipation member such as a heat dissipation sheet.

[0003] However, in each of Japanese Laid-Open Patent Publication (kokai) No. 2015-115417 and Japanese Laid-Open Patent Publication (kokai) No. 2001-77569, since grease is used for the liquid heat dissipation member, there is a problem that the liquid heat dissipation member leaks or adheres to gloves or work clothes depending on the skill level of an operator, for example, at the time of assembly or disassembly work. A heat dissipation gap filler may be used instead of the grease. The “heat dissipation gap filler” is a paste-like filler containing a filler having thermal conductivity, and can be cured with time after being applied to a predetermined portion as a cooling structure. Then, the heat dissipation gap filler in a cured state has extremely strong adhesive force to the predetermined portion. Therefore, for example, even if the heat dissipation gap filler in the cured state is tried to be peeled off from the predetermined portion at the time of disassembly work, it is difficult to peel off the heat dissipation gap filler, and there arises a problem that workability is deteriorated.

SUMMARY OF THE INVENTION

[0004] The present invention provides an electronic device excellent in disassembly workability at the time of repair or the like while sufficiently ensuring thermal conductivity for transferring heat from a heating element to a cooling duct.

[0005] According to an aspect of the invention, the present invention provides an electronic device comprising a circuit board having at least one heating element that generates heat by energization, a cooling duct disposed to face the circuit board and through which air for cooling the heating element passes, a heat dissipation filler provided in contact with a surface of the cooling duct facing the circuit board and having thermal conductivity for transferring heat from the heating element to the cooling duct, a flexible member provided between the circuit board and the heat dissipation filler and having flexibility, and a positioning unit configured to position the flexible member at a position where contact between the circuit board and the heat dissipation filler can be prevented.

[0006] According to the present invention, it is possible to achieve excellent disassembly workability at the time of repair or the like while sufficiently ensuring the thermal conductivity for

transferring heat from the heating element to the cooling duct.

[0007] Further features of the present invention will become apparent from the

[0008] following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is an external perspective view of an image pickup apparatus to which an electronic device according to a first embodiment is applied as viewed from an upper front.

[0010] FIG. 2 is an external perspective view of the image pickup apparatus to which the electronic device according to the first embodiment is applied as viewed from an upper rear.

[0011] FIG. 3 is an exploded perspective view showing an internal configuration of the image pickup apparatus.

[0012] FIG. 4 is a plan view of a forced air cooling structure in an internal structure.

[0013] FIGS. 5A and 5B are plan views of the forced air cooling structure in the internal structure.

[0014] FIG. 6 is an exploded perspective view of a main unit of the image pickup apparatus.

[0015] FIG. 7 is a plan view of a control circuit board.

[0016] FIG. 8 is a diagram showing a positional relationship in an assembled state of the control circuit board, a film, a heat dissipation gap filler, and a forced air cooling duct.

[0017] FIG. 9 is a front view of the film and the forced air cooling duct as viewed from a front surface side of the image pickup apparatus.

[0018] FIG. 10 is a rear view of the control circuit board and the film as viewed from a rear surface side of the image pickup apparatus.

[0019] FIG. 11 is a front view of the forced air cooling duct of the image pickup apparatus to which the electronic device according to a second embodiment is applied as viewed from the front surface side of the image pickup apparatus.

[0020] FIG. 12 is a diagram showing the positional relationship in the assembled state of the control circuit board, the film, the heat dissipation gap filler, and the forced air cooling duct.

[0021] FIG. 13 is a front view of the forced air cooling duct of the image pickup apparatus to which the electronic device according to a third embodiment is applied as viewed from the front surface side of the image pickup apparatus.

[0022] FIG. 14 is a sectional view taken along line B-B in FIG. 13.

DESCRIPTION OF THE EMBODIMENTS

[0023] Hereinafter, embodiments of the present invention will be described in detail with reference to the drawings. However, configurations described in the following embodiments are merely examples, and the scope of the present invention is not limited by the configurations described in the embodiments. For example, each of parts constituting the present invention can be replaced with a unit having any configuration capable of exhibiting a similar function. Further, an arbitrary component may be added. Furthermore, any two or more configurations (features) of the embodiments may be combined.

[0024] Hereinafter, a first embodiment will be described with reference to FIGS. 1 to 10.

External Configuration of Image Pickup Apparatus

[0025] FIG. 1 is an external perspective view of an image pickup apparatus to which an electronic device according to the first embodiment is applied as viewed from an upper front. FIG. 2 is an external perspective view of the image pickup apparatus to which the electronic device according to the first embodiment is applied as viewed from an upper rear. It should be noted that as shown in FIGS. 1 and 2, in the present embodiment, an X axis, a Y axis, and a Z axis orthogonal to each other are set (the same applies to FIG. 3 and subsequent figures). The Z axis is an axis set parallel to a front-rear direction of an image pickup apparatus 100, that is, an axis perpendicular to an

image pickup surface **101**. In the Z axis, a direction from the rear surface side to the front surface side of the image pickup apparatus **100** is defined as a positive direction, and the opposite direction is defined as a negative direction. The Y axis is an axis set in parallel with an up-down direction of the image pickup apparatus **100**. In the Y axis, a direction from a lower surface to an upper surface of the image pickup apparatus **100** is defined as a positive direction, and the opposite direction is defined as a negative direction. The X axis is an axis set in parallel with a left-right direction of the image pickup apparatus **100**. In the X axis, a direction from a left side surface to a right side surface of the image pickup apparatus **100** when the image pickup apparatus **100** is viewed from the front is defined as a positive direction, and the opposite direction is defined as a negative direction. The image pickup apparatus **100** has an image pickup function of shooting a moving image or a still image, and is an apparatus to which the electronic device of the present invention is applied.

[0026] As shown in FIG. 1, the image pickup apparatus **100** includes a lens attachment part **102** provided on a front surface, and an image pickup surface **101** provided on the negative side in the Z-axis direction with respect to the lens attachment part **102**, that is, on the rear side with respect to the lens attachment part **102**. A lens barrel (not shown) including plural types of lenses having different optical performances is detachably attached to the lens attachment part **102**. The lens barrel is communicably connected to the image pickup apparatus **100** in a state of being attached to the lens attachment part **102**. Thus, each of the lenses can be driven. Further, the lens barrel forms an image on the image pickup surface **101** by each of the lenses in the state of being attached to the lens attachment part **102**. The image pickup surface **101** generates video data based on an optical image formed on the image pickup surface **101**. Further, the image pickup apparatus **100** includes an operation member **103** provided on the right side surface and an accessory mounting electrical contact **104** provided on an upper surface. The operation member **103** includes, for example, a plurality of buttons, dials, and the like. By operating the operation member **103**, for example, it is possible to switch ON/OFF of a power supply of the image pickup apparatus **100**, start/stop image pickup, adjust an image and sound, and the like. The accessory mounting electrical contact **104** communicably connects an external accessory detachably mounted on the upper side of the image pickup apparatus **100** and the image pickup apparatus **100**. Thus, the external accessory can be controlled. The external accessory is not particularly limited, and examples thereof include a microphone, a lighting device, and a handle. As shown in FIG. 2, the image pickup apparatus **100** includes a general-purpose accessory attachment part **105** and a speaker **106** provided on the rear surface. A general-purpose accessory can be attached to the general-purpose accessory attachment part **105**. The general-purpose accessory is not particularly limited, and examples thereof include a tripod. The speaker **106** emits out the sound of the moving image, for example.

Internal Configuration of Image Pickup Apparatus

[0027] FIG. 3 is an exploded perspective view showing an internal configuration of the image pickup apparatus. As shown in FIG. 3, the image pickup apparatus **100** has an internal structure **300**. Internal structure **300** includes an image pickup device **310**, a sensor duct unit **520**, a control circuit board (circuit board) **501**, a forced air cooling duct (cooling duct) **505**, a power supply board **350**, and a cooling fan (fan) **506**. Further, the internal structure **300** includes a media duct unit **380**, a media substrate **390**, a sub-media substrate **400**, and a wireless substrate **312**. These members constituting the internal structure **300** are arranged from the positive side to the negative side in an optical axis direction, that is, the Z-axis direction. Further, the members adjacent to each other in the Z-axis direction are arranged to face each other. The image pickup device **310** has the image pickup surface **101** that converts light transmitted through each of the lenses into an electric signal. The control circuit board **501** is a board that controls the entire image pickup apparatus **100**. The control circuit board **501** is a board on which a plurality of elements is mounted and which has the largest area in the internal structure **300**. The element executed on the control circuit board **501** is not particularly limited, and examples thereof include an element for processing a signal from the

image pickup device **310**, an element for executing processing such as color adjustment of a video, and a memory used for these elements. The power supply board **350** constitutes a power supply circuit that supplies power to electric components in the image pickup apparatus **100** together with the control circuit board **501**. The media substrate **390** is a substrate for recording a main video on a recording medium. The sub-media substrate **400** is a substrate that stores a setting state of an image pickup condition and records a backup video in which a data capacity of the main video recorded on the media substrate **390** is reduced. The media duct unit **380** is provided with an opening **382** penetrating in the Z-axis direction. The media substrate **390** is configured so that at least a part thereof is exposed through the opening **382**. The wireless substrate **312** is electrically connected to a wireless antenna **311** and controls wireless communication with an external device. The wireless antenna **311** is disposed in an upper rear part of the internal structure **300**.

Forced Air Cooling Structure in Internal Structure

[0028] FIGS. **4**, **5A**, and **5B** are each plan views of the forced air cooling structure in the internal structure. It should be noted that FIG. **4** is a plan view of the forced air cooling structure in the internal structure as viewed from the positive side in the Y-axis direction. FIG. **5A** is a plan view of the forced air-cooling structure in the internal structure as viewed from the negative side in the Z-axis direction. FIG. **5B** is a sectional view taken along line A-A in FIG. **5A**. As shown in FIG. **4**, the forced air cooling structure of the internal structure **300** includes a heat dissipation duct **301** and the cooling fan **506**. As described above, the plurality of elements is mounted on the control circuit board **501**. These elements are heating elements (heating elements **610** to be described later) that generate heat by energization. The forced air cooling structure is configured to be able to release heat from each element, that is, each heating element. It should be noted that the number of elements mounted on the control circuit board **501** is plural in the present embodiment, but is not limited thereto, and may be at least one. Further, in the present embodiment, each element is provided on a surface (surface **501a** to be described later) of the control circuit board **501** facing the forced air cooling duct **505**, that is, a surface facing the negative side in the Z-axis direction, but is not limited thereto. For example, the elements may be provided on a surface of the control circuit board **501** facing the positive side in the Z-axis direction, or may be provided in a dispersed manner on the surface of the control circuit board **501** facing the positive side in the Z-axis direction and a surface of the control circuit board **501** facing the negative side in the Z-axis direction. The heat dissipation duct **301** includes the sensor duct unit **520**, the forced air cooling duct **505**, the media duct unit **380**, and an exhaust duct unit **370**.

[0029] The sensor duct unit **520** has a sensor duct intake port **521** that opens to the negative side in the X-axis direction. Air (Outside air) AR is sucked into the sensor duct intake port **521**. The sensor duct unit **520** is coupled to the forced air cooling duct **505** via a sensor duct coupling part **542**. Thus, the sensor duct unit **520** and the forced air cooling duct **505** communicate with each other, so that air AR1 sucked into the sensor duct intake port **521** can sequentially pass through the sensor duct unit **520** and the forced air cooling duct **505**. It should be noted that the heat generated in the image pickup device **310** is transmitted to the sensor duct unit **520** via a heat conduction member (not shown) such as a graphite sheet. Then, the heat is cooled as the air AR1 passes through the sensor duct unit **520**.

[0030] As shown in FIG. **5B**, the media duct unit **380** has a media duct intake port **581** opened to the positive side in the X-axis direction. Air AR2 is sucked into the media duct intake port **581**. The media duct unit **380** is coupled to the forced air cooling duct **505** via a media duct coupling part **543**. Thus, the media duct unit **380** and the forced air cooling duct **505** communicate with each other, so that the air AR2 sucked into the media duct intake port **581** can sequentially pass through the media duct unit **380** and the forced air cooling duct **505**. As described above, at least a part of the media substrate **390** is exposed through the opening **382** of the media duct unit **380**. Thus, the heat generated in the media substrate **390** is cooled as the air AR2 passes through the media duct unit **380**. The sub-media substrate **400** is disposed on the negative side in the Z-axis direction with

respect to the media substrate **390**. Since the sub-media substrate **400** is a substrate on which small data is written, and thus an amount of heat generation is also relatively small. Therefore, forced cooling of the sub-media substrate **400** may be omitted.

[0031] The forced air cooling duct **505** has a forced air cooling duct intake port **541** that opens to the positive side in the X-axis direction. Air AR3 is sucked into the forced air cooling duct intake port **541**. Thus, the air AR3 can pass through the forced air cooling duct **505**. It should be noted that the heat generated in the control circuit board **501** (in particular, the heating element) is cooled as the air AR3 passes through the forced air cooling duct **505**. A part of the heat generated in the control circuit board **501** is also cooled in the sensor duct unit **520**. The cooling fan **506** is coupled to the forced air cooling duct **505** via a cooling fan coupling hole **544**. This places the cooling fan **506** and the forced air cooling duct **505** in communication with each other, and thus when the cooling fan **506** operates, the air AR1 to the air AR3 are forced to pass to the forced air cooling duct **505**. The cooling fan **506** has the exhaust duct unit **370** that opens to the positive side in the X-axis direction. The air AR1 to the air AR3 having passed through the forced air cooling duct **505** are discharged to an outside of the image pickup apparatus **100** via the exhaust duct unit **370** by operation of the cooling fan **506**.

Configuration of Main Unit

[0032] FIG. **6** is an exploded perspective view of a main unit of the image pickup apparatus. As shown in FIG. **6**, the image pickup apparatus **100** has a main unit **500**. The main unit **500** includes the control circuit board **501**, a film (flexible member) **503**, a heat dissipation gap filler (heat conduction member) **504**, the forced air cooling duct **505**, and the cooling fan **506**. These members constituting the main unit **500** are arranged in order from the positive side to the negative side in a normal direction of the control circuit board **501**, that is, the Z-axis direction. Therefore, among these members, the control circuit board **501** is located closest to a subject side. The forced air cooling duct **505** is a flat member disposed to face the control circuit board **501**. The forced air cooling duct **505** is made of a metal material having relatively high thermal conductivity, such as aluminum, and is thermally connected to the heating element **610** mounted on the control circuit board **501**. As described above, the heating element **610** is an element that generates heat by energization. The heat generated by the heating element **610** is heat-exchanged with the air AR2 sucked into the forced air cooling duct **505** by rotation of the cooling fan **506**. Thus, the heating element **610** can be forcibly cooled.

[0033] The heat dissipation gap filler (heat dissipation filler) **504** is disposed between the heating element **610** and the forced air cooling duct **505**. The heat dissipation gap filler **504** is a heat conduction member that transfers heat from the heating element **610** to the forced air cooling duct **505**. In the present embodiment, the heat dissipation gap filler **504** is provided in contact with a surface **505a** of the forced air cooling duct **505** facing the control circuit board **501**, that is, the surface **505a** facing the positive side in the Z-axis direction. The heat dissipation gap filler **504** has a paste shape, and is cured over time after being applied to the surface **505a**. Then, the heat dissipation gap filler **504** in a cured state has extremely strong adhesive force to the surface **505a**, and is difficult to peel off from the surface **505a**. It should be noted that the shape of the heat dissipation gap filler **504** in the cured state is a flat shape in configuration shown in FIG. **6**, but is not limited thereto. Further, the heat dissipation gap filler **504** is not particularly limited, and for example, a material containing ceramic fillers having thermal conductivity and a binder including a silicone-based resin that bonds the fillers can be used.

[0034] Here, in the image pickup apparatus **100**, it is conceivable to use a heat dissipation sheet having flexibility instead of the heat dissipation gap filler **504**. In general, the heat dissipation sheet is used in a state of being sandwiched between members to be heat-exchanged. However, depending on various conditions such as a material constituting the heat dissipation sheet and a thickness of the heat dissipation sheet, there is a possibility that a force for deforming each member is generated from the heat dissipation sheet on each member. In this case, the control circuit board

501 may be deformed such as bent by the force from the heat dissipation sheet, resulting in a defect or failure. In contrast, since the heat dissipation gap filler **504** is in the cured state, the above concern can be eliminated as compared with the heat dissipation sheet. It is also conceivable to use heat dissipation grease instead of the heat dissipation gap filler **504**. Unlike the heat dissipation gap filler **504**, the heat dissipation grease is not in the cured state and remains as liquid. Thus, the heat dissipation gap filler **504** can prevent, for example, pumping out due to contraction and expansion by heat and leakage due to vibration as compared with the heat dissipation grease.

Arrangement of Heating Element (Element) on Control Circuit Board

[0035] FIG. 7 is a plan view of the control circuit board. As shown in FIG. 7, a heating element **601a**, a heating element **601b**, and a heating element **601c** are mounted as the heating element **610** on the surface **501a** facing the negative side in the Z-axis direction of the control circuit board **501**. The heating elements **601a** to **601c** are arranged at intervals in the X-axis direction. Each of the heating elements **601a** to **601c** is, for example, a CPU. It should be noted that shapes and sizes of the heating elements **601a** to **601c** are not limited to those of configuration shown in FIG. 7. The number of heating elements constituting the heating element **610** is 3 in the configuration shown in FIG. 7, but is not limited thereto, and may be, for example, 1, 2, or 4 or more. On the surface **501a** of the control circuit board **501**, a conductive member **620a**, a conductive member **620b**, and a conductive member **620c** are mounted. Although the conductive members **620a** to **620c** are arranged at intervals, arrangement positions thereof are not particularly limited. Each of the conductive members **620a** to **620c** projects from the surface **501a** and functions as a connection part electrically connected to the forced air cooling duct **505**. It should be noted that shapes and sizes of the conductive members **620a** to **620c** are not limited to those of the configuration shown in FIG. 7. In addition, the number of conductive members is three in the configuration shown in FIG. 7, but is not limited thereto, and may be, for example, one, two, or four or more.

Relationship Between Heat Dissipation Gap Filler and Film

[0036] FIG. 8 is a diagram showing a positional relationship in an assembled state of the control circuit board, the film, the heat dissipation gap filler, and the forced air cooling duct. As shown in FIG. 8, the film **503** is disposed between the heating element **610** of the control circuit board **501** and the heat dissipation gap filler **504**. The film **503** is a sheet-like member having flexibility. The constituent material of the film **503** is not particularly limited, and is, for example, a resin material such as polyester. The thickness of the film **503** is preferably, for example, several μm to several tens mm. When heat is generated from the heating element **610**, the heat can be quickly transmitted to the forced air cooling duct **505** through the film **503** and the heat dissipation gap filler **504** in this order. As described above, the film **503** can sufficiently ensure the thermal conductivity.

[0037] The film **503** is sandwiched between the heating element **610** and the heat dissipation gap filler **504**. Thus, it is possible to prevent the heating element **610** and the heat dissipation gap filler **504** from coming into close contact with each other (coming into contact with each other). Then, due to this adhesion preventing effect, when the main unit **500** is disassembled, for example, to repair the control circuit board **501** or to replace the heating element **610**, the forced air cooling duct **505** can be easily separated from the control circuit board **501** together with the heat dissipation gap filler **504**. Thus, it is possible to omit an operation in which a part of the heat dissipation gap filler **504** remains on the control circuit board **501** and the heat dissipation gap filler **504** is removed from the control circuit board **501**. As a result, repair of the control circuit board **501**, replacement of the heating element **610**, and the like can be easily performed. Since the film **503** is disposed between the heating element **610** of the control circuit board **501** and the heat dissipation gap filler **504** in an assembled state of the main unit **500** as described above, disassembly workability at the time of repair or the like is excellent. It should be noted that in a disassembled state of the main unit **500**, the film **503** directly adheres to the heat dissipation gap filler **504** or is peeled off from the heat dissipation gap filler **504**. When the film **503** directly adheres to the heat dissipation gap filler **504**, the film **503** may be peeled off from the heat

dissipation gap filler **504** and replaced with a new film **503**. As described above, the film **503** has flexibility, and the thickness is preferably in the above numerical range. Due to combination of the flexibility and the numerical range, the film **503** can easily and sufficiently follow and adhere to the heating element **610** and the heat dissipation gap filler **504** regardless of the number, arrangement positions, and surface shapes of the heating elements **610** and the heat dissipation gap fillers **504**.

Positioning of Film

[0038] As described above, the film **503** is a member capable of exhibiting the adhesion preventing effect of preventing the heating element **610** of the control circuit board **501** and the heat dissipation gap filler **504** from adhering to each other. However, since the film **503** is made of a sheet material having flexibility, positional displacement tends to easily occur, and when the positional displacement occurs, the adhesion preventing effect may not be sufficiently exhibited. Therefore, the main unit **500** (image pickup apparatus **100**) is configured so that film **503** can be positioned, that is, positional displacement can be prevented. Hereinafter, this configuration and operation will be described.

[0039] FIG. **9** is a front view of the film and the forced air cooling duct as viewed from the front surface side of the image pickup apparatus. As shown in FIG. **9**, the main unit **500** has a positioning unit **800** that positions the film **503**. The positioning unit **800** is configured to position the film **503** at a position where close contact between the heating element **610** of the control circuit board **501** and the heat dissipation gap filler **504** can be prevented. Hereinafter, a state in which the film **503** is positioned may be referred to as a “positioning state”. In the present embodiment, the positioning unit **800** has a through-hole (positioning hole) **801** provided to penetrate the film **503**, and a projecting part **802** provided to project toward the positive side in the Z axis direction on the forced air cooling duct **505**. The projecting part **802** is constituted by, for example, a boss (pin) screwed or press-fitted into forced air cooling duct **505**, and passes through through-hole **801**. Two through-holes **801** and two projecting parts **802** are provided. When viewed from the Z-axis direction, the two through-holes **801** are arranged on opposite sides to each other with the heating element **610** interposed therebetween (see FIGS. **8** and **9**). On the other hand, similarly to the through-holes **801**, the two projecting parts **802** are also arranged on opposite sides to each other with the heating element **610** interposed therebetween (see FIGS. **8** and **9**). Thus, two positioning positions for the film **503** can be arranged as far apart as possible. It should be noted that in positioning, it is preferable to arrange the two positioning positions as far apart as possible.

[0040] With the positioning unit **800** having such a configuration, the position of the film **503** in the X-axis direction and the Y-axis direction is regulated, and the film is reliably positioned at a position where the heating element **610** and the heat dissipation gap filler **504** can be prevented from coming into close contact with each other. This prevents the positional displacement of the film **503**, and thus a state in which the adhesion preventing effect by the film **503** is exhibited is maintained. As described above, the film **503** is sandwiched between the heating element **610** and the heat dissipation gap filler **504**. This regulates the position of the film **503** in the Z-axis direction. Due to combination of this position regulation in the Z-axis direction and position regulation in the X-axis direction and the Y-axis direction by the positioning unit **800**, the positioning state of the film **503** is more reliably maintained.

[0041] The projecting part **802** has a columnar shape. The through-hole **801** has a circular shape, and its radius is larger than that of the projecting part **802** by a clearance C. Therefore, in the present embodiment, the columnar projecting part **802** and the circular through-hole **801** are in a clearance fit relationship. This improves followability (shape followability) of the film **503** with respect to the heating element **610** and the heat dissipation gap filler **504** described above. It should be noted that the number of the through-holes **801** and the projecting parts **802** formed is two in the present embodiment, but is not limited thereto, and may be at least two. In the present embodiment, the projecting part **802** is provided on the forced air cooling duct **505**, but is not limited thereto, and for example, may be provided on the control circuit board **501**. In the present embodiment, the

positioning unit **800** is configured to have the through-hole **801** and the projecting part **802**, but is not limited thereto, and for example, may be provided in a part of the main unit **500** and may be constituted by a portion that can be positioned by abutting an edge of the film **503**.

[0042] FIG. **10** is a rear view of the control circuit board and the film as viewed from the rear surface side of the image pickup apparatus. As shown in FIG. **10**, the film **503** includes a relief **820a** for avoiding interference with the conductive member **620a** of the control circuit board **501** in the positioning state, a relief **820b** for avoiding interference with the conductive member **620b**, and a relief **820c** for avoiding interference with the conductive member **620c**. Each of the relief **820a** and the relief **820b** is formed to penetrate the film **503**, and the relief **820c** is formed by the edge of the film **503** entering toward the center side of the film **503**. Due to such reliefs **820a** to **820c**, the conductive members **620a** to **620c** are exposed toward the forced air cooling duct **505**. Thus, the conductive members **620a** to **620c** can be electrically connected to the forced air cooling duct **505**. The conductive member **620a** and the conductive member **620c** also function as vent holes through which the air passes. This improves heat dissipation. In addition, since deformation is facilitated in peripheral portions of the conductive member **620a** and the conductive member **620c**, the followability of the film **503** with respect to the heating element **610** and the heat dissipation gap filler **504** described above is improved.

[0043] Hereinafter, a second embodiment will be described with reference to FIGS. **11** and **12**, but differences from the above-described embodiment will be mainly described, and description of similar matters will be omitted. FIG. **11** is a front view of the forced air cooling duct of the image pickup apparatus to which the electronic device according to the second embodiment is applied as viewed from the front surface side of the image pickup apparatus. FIG. **12** is a diagram showing the positional relationship in the assembled state of the control circuit board, the film, the heat dissipation gap filler, and the forced air cooling duct.

[0044] As shown in FIG. **11**, the forced air cooling duct **505** has a convex shape **900** (a portion surrounded by an alternate long and short dash line in FIG. **11**) projecting toward the negative side in the Z-axis direction. As shown in FIG. **12**, the convex shape **900** is formed to face the heating element **610** with the heat dissipation gap filler **504** and the film **503** interposed therebetween. Thus, the convex shape **900** functions as a maintaining unit that maintains a constant distance D between the heating element **610** and the forced air cooling duct **505** in the Z-axis direction. By maintaining the distance D constant, it is possible to prevent a difference in heat transfer performance to the forced air cooling duct **505** among the heating elements **610**.

[0045] Hereinafter, a third embodiment will be described with reference to FIGS. **13** and **14**, but differences from the above-described embodiment will be mainly described, and description of similar matters will be omitted. FIG. **13** is a front view of the forced air cooling duct of the image pickup apparatus to which the electronic device according to the third embodiment is applied as viewed from the front surface side of the image pickup apparatus. The forced air cooling duct **505** has a heat dissipation rib (rib) **1004** provided to project from a surface **505a** facing the positive side in the Z-axis direction. The heat dissipation rib **1004** is formed integrally with the forced air cooling duct **505**, and has a shape surrounding the heat dissipation gap filler **504** when viewed from the Z-axis direction. FIG. **14** is a sectional view taken along line B-B in FIG. **13**. As shown in FIG. **14**, the heat dissipation rib **1004** surrounds the heating element **610** together with the heat dissipation gap filler **504**. Thus, the film **503** is disposed between the heat dissipation gap filler **504** and the heating element **610**, and can also be in contact with a side surface of the heating element **610**. A contact area of the film **503** with the heating element **610** increases by an amount of contact with the side surface of the heating element **610**. Thus, the heat from the heating element **610** is quickly transferred to the forced air cooling duct **505** via the film **503**. A distance between the side surface of the heating element **610** and the heat dissipation rib **1004** is preferably $\frac{1}{2}$ or less of the thickness of the film **503**. Thus, the film **503** is compressed by the heat dissipation rib **1004** and can be reliably brought into contact with the side surface of the heating element **610**.

Other Embodiments

[0046] Although the preferred embodiments of the present invention have been described above, the present invention is not limited to the above-described embodiments, and various modifications and changes can be made within the scope of the gist of the present invention. It should be noted that as an apparatus to which the electronic device of the present invention is applicable, the image pickup apparatus **100** is described in each of the above embodiments, but the present invention is not limited thereto, and for example, an information processing apparatus such as a desktop or notebook personal computer, a tablet terminal, or a smartphone may be used.

[0047] This application claims the benefit of Japanese Patent Application No. 2024-032030, filed Mar. 4, 2024 which is hereby incorporated by reference wherein in its entirety.

Claims

1. An electronic device comprising: a circuit board having at least one heating element that generates heat by energization; a cooling duct disposed to face the circuit board and through which air for cooling the heating element passes; a heat dissipation filler provided in contact with a surface of the cooling duct facing the circuit board and having thermal conductivity for transferring heat from the heating element to the cooling duct; a flexible member provided between the circuit board and the heat dissipation filler and having flexibility; and a positioning unit configured to position the flexible member at a position where contact between the circuit board and the heat dissipation filler can be prevented.
2. The electronic device according to claim 1, wherein the positioning unit includes a through-hole provided to penetrate the flexible member, and a projecting part provided to project from the cooling duct or the circuit board and passing through the through-hole.
3. The electronic device according to claim 2, wherein at least two through-holes and at least two projecting parts are provided.
4. The electronic device according to claim 3, wherein when viewed from a normal direction of the circuit board, the two through-holes are arranged on opposite sides to each other with the heating element interposed therebetween, and the two projecting parts are also arranged on opposite sides to each other with the heating element interposed therebetween.
5. The electronic device according to claim 2, wherein the through-hole has a circular shape, and the projecting part has a columnar shape having a clearance fit relationship with the through-hole.
6. The electronic device according to claim 1, wherein the circuit board includes a plurality of the heating elements, and the positioning unit positions the flexible member so as to be able to prevent contact between each of the heating elements and the heat dissipation filler.
7. The electronic device according to claim 1, wherein the heating element is provided on a surface of the circuit board facing the cooling duct, and the flexible member is sandwiched between the heating element and the heat dissipation filler.
8. The electronic device according to claim 1, wherein the circuit board includes a connection part provided to project from a surface facing the cooling duct and is electrically connected to the cooling duct, and the flexible member includes a relief that avoids interference with the connection part.
9. The electronic device according to claim 1, wherein the flexible member has a sheet shape.
10. The electronic device according to claim 1, wherein the electronic device includes a fan that forcibly allows the air to pass through the cooling duct.
11. The electronic device according to claim 10, wherein when viewed from a normal direction of the circuit board, the circuit board, the flexible member, the heat dissipation filler, the cooling duct, and the fan are arranged in this order along the normal line.
12. The electronic device according to claim 11, wherein the electronic device is an image pickup apparatus having an image pickup function, and among the circuit board, the flexible member, the

heat dissipation filler, the cooling duct, and the fan, the circuit board is located closest to a subject side.

13. The electronic device according to claim 1, wherein the circuit board includes a plurality of the heating elements provided on a surface facing the cooling duct, and the electronic device includes a maintaining unit that maintains a constant distance between the heating elements and the cooling duct in a normal direction of the circuit board.

14. The electronic device according to claim 1, wherein the cooling duct includes a rib provided to project from a surface facing the circuit board and having a shape surrounding the heating element when viewed from a normal direction of the circuit board.

15. The electronic device according to claim 1, wherein the electronic device is an image pickup apparatus having an image pickup function.
